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[Editor name], [Title]
Clinical Trials

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Dear [Editor name],

We wish to submit an original research article entitled “Sampling Design for Spatial Cluster Randomized Trials Under Heterogeneous Incidence” for consideration by *Clinical Trials*. We confirm that this work is original and has not been published elsewhere, nor is it currently under consideration for publication elsewhere. The study extends our recent work in *BMC Medical Research Methodology* (DOI 10.1186/s12874-026-02968-0) on how the treatment assignment design of a spatial cluster randomized trial shapes estimation when spillover effects and spatial dependence are present.

In this paper, we report a simulation study that compares six treatment assignment designs and simple random sampling (SRS) on a 100-cluster grid under heterogeneous baseline incidence, spatial dependence and two spillover regimes. We pair it with a completed application that uses observed 2018–2021 sudden unexpected death (SUD) incidence for 58 North Carolina community-college service areas to plan an education pilot for healthcare professionals. Because a trial realizes only one allocation, we evaluate each design by both its average mean squared error (MSE) and the spread of error across the allocations it can produce.

Our findings show that the best design depends on where spillover is expected to reach. When spillover reaches mainly control clusters, saturation designs, which treat different proportions of clusters in different regions, reduced mean MSE by roughly 21–30% relative to SRS in the application and also lowered the risk of a poor allocation; incidence-guided saturation achieved this while directing the highest treatment saturations to the highest-burden regions. When spillover reaches both arms, SRS remained competitive. We believe this manuscript is appropriate for publication by *Clinical Trials* because it gives investigators practical, scenario-specific guidance for choosing an allocation strategy in spatial cluster randomized trials, demonstrated on a real planning problem, while transparently reporting its limitations, including interval undercoverage and unequal treatment budgets.

We have no conflicts of interest to disclose, and this research received no specific funding.

Please address all correspondence concerning this manuscript to the corresponding author, Dr. Feng-Chang Lin, at flin33@email.unc.edu or (919) 966-9431, and to me at awalther@unc.edu.

Thank you for your consideration of this manuscript.

Sincerely,

Andrew Walther